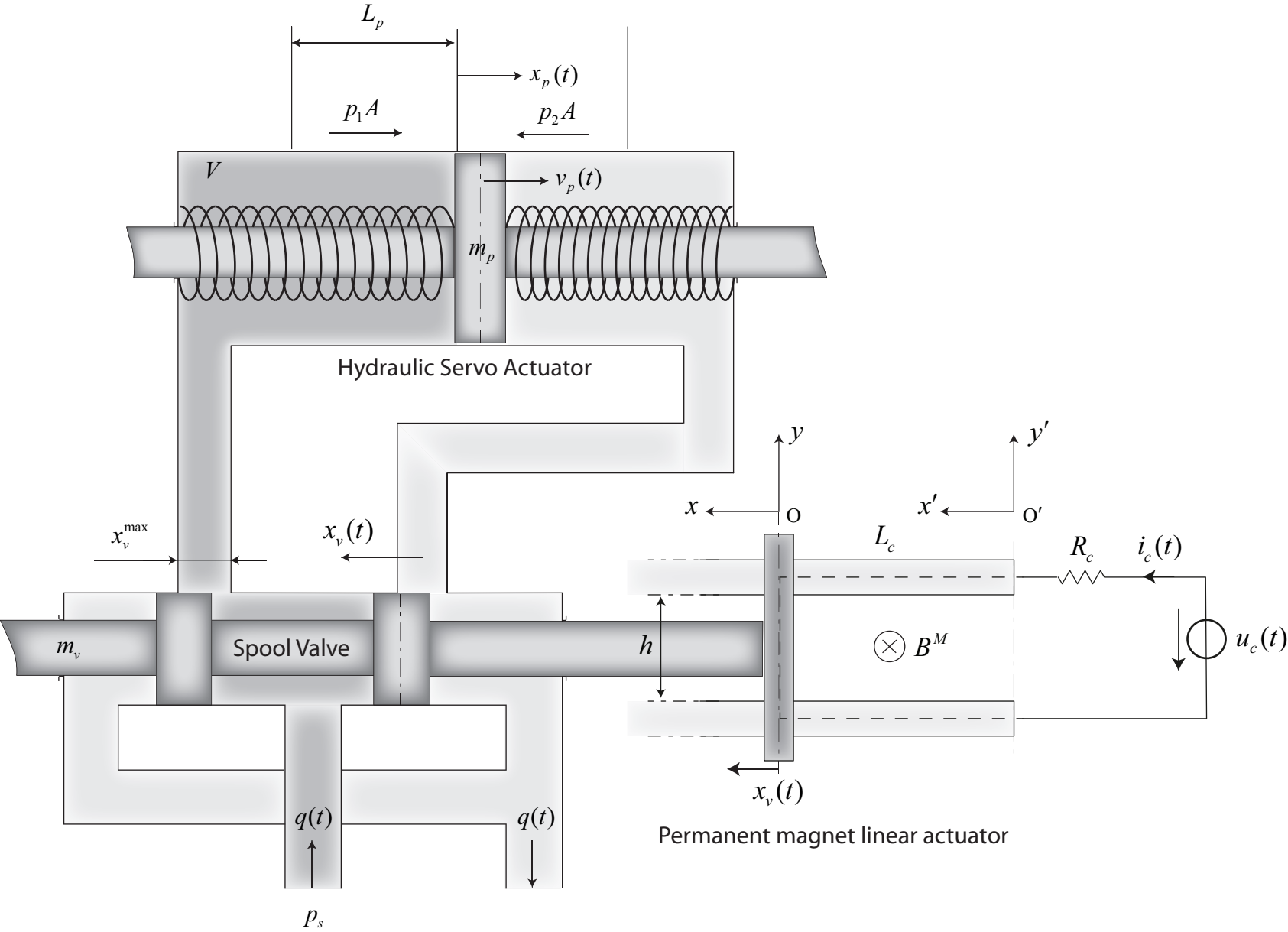

L_p
$x_p(t)$
$p_1 A$
$p_2 A$
V
$v_p(t)$
m_p
Hydraulic Servo Actuator
y
y'
x
O
x'
O′
$x_v^{\max}$
$x_v(t)$
L_c
R_c
$i_c(t)$
m_v
Spool Valve
h
B^M
$u_c(t)$
$x_v(t)$
$q(t)$
$q(t)$
p_s
Permanent magnet linear actuator